

ML Scan Parameters

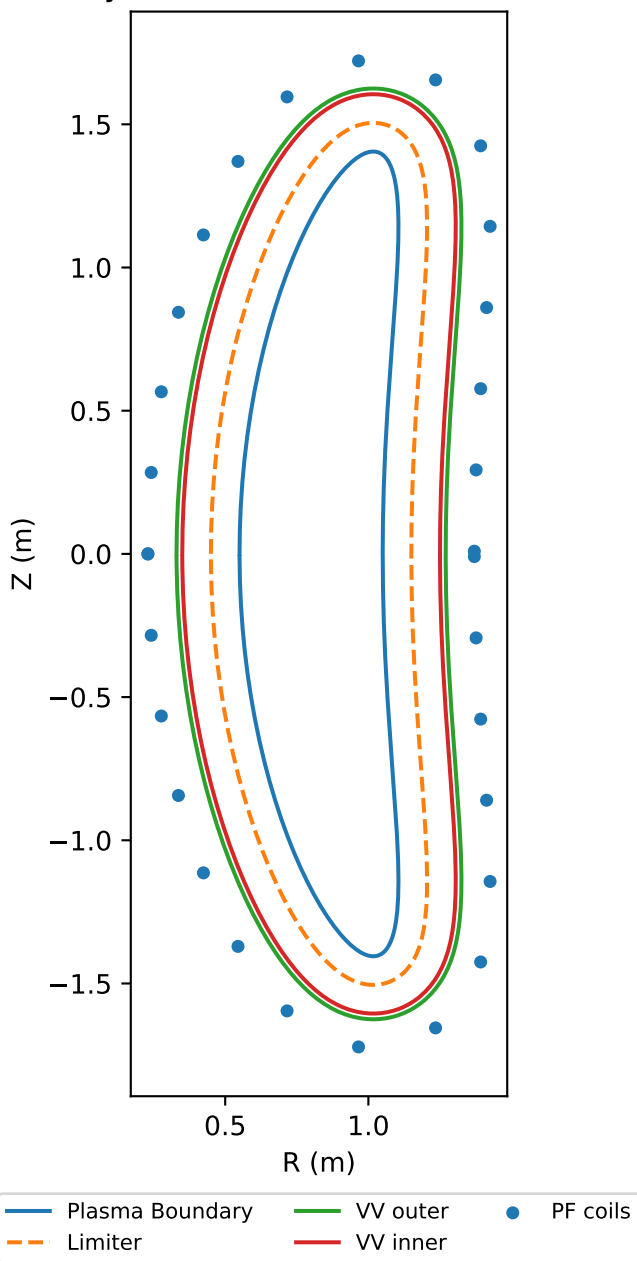
C-shape plasma $(r,z) = (A + B*\sin t + C*\cos 2t, F + G*\cos t - H*\sin 2t)$

A=0.9200 B=0.2500 C=0.1200
G=1.4000 H=0.0600 F=0.0
R0=0.8000 m kappa_target=5.600 delta=-0.060
B0=1.8 T Ip=1.00 MA

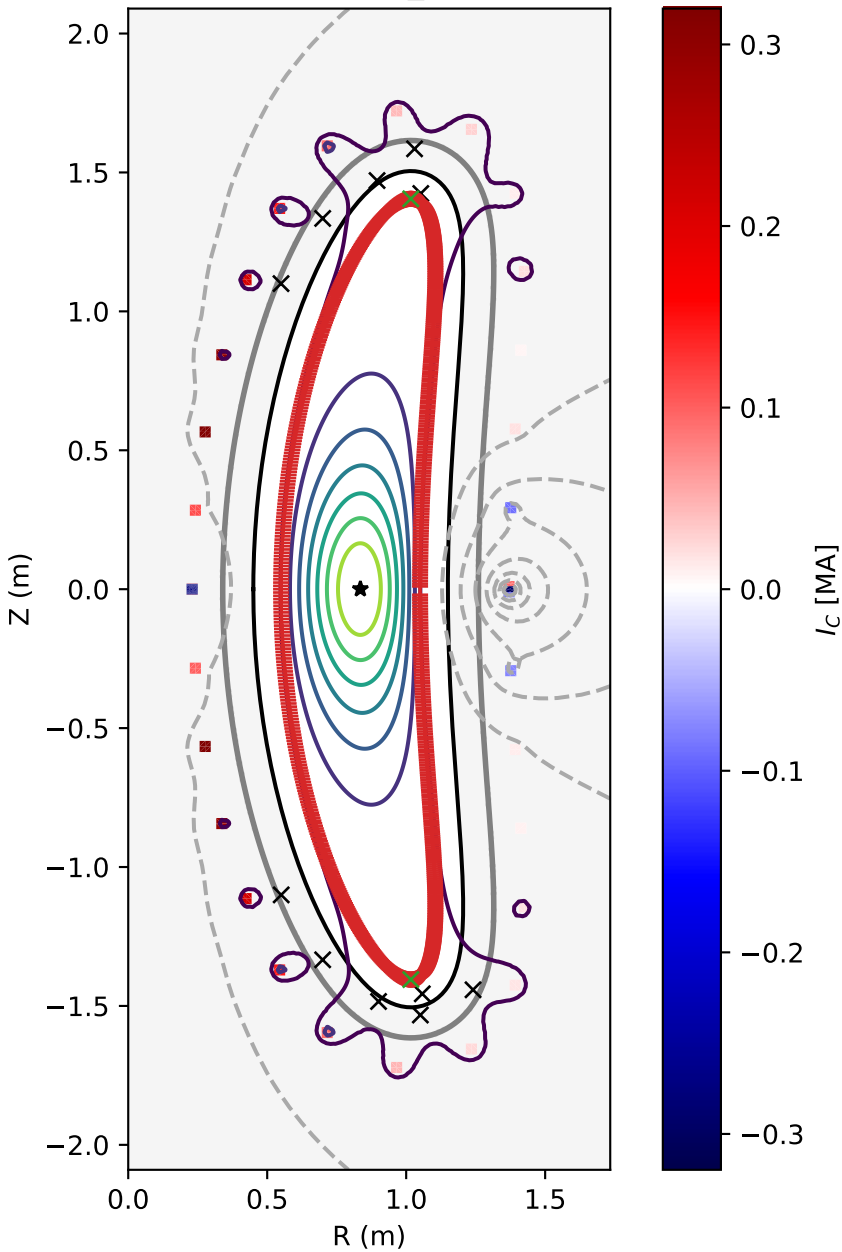
Plasma profiles:
pp_prof: $(1 - \psi^{0.743})^{1.761}$
ffp_prof: $(1 - \psi^{0.983})^{0.625}$

Scan params: B=2.5 C=12 G=14 H=6
alpha_pp=0.743 gamma_pp=1.761
alpha_ff=0.983 gamma_ff=0.625

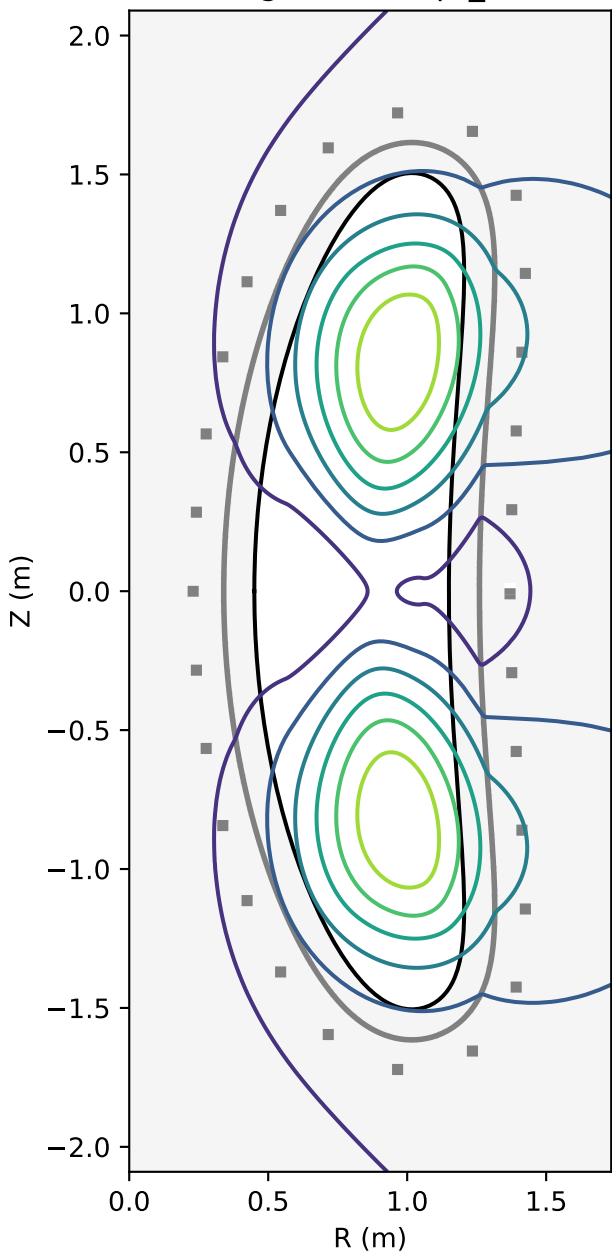
Geometry B=2.5, C=12, G=14, H=6, $\kappa=5.6$



Equilibrium flux $\kappa_{\text{actual}}=4.96$



Unstable eigenmode $\gamma\tau_w = -1.000$



Vertical Stability Analysis

Vertical Stability Results

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B=2.5 C=12 G=14 H=6

alpha_pp=0.743 gamma_pp=1.761

alpha_ff=0.983 gamma_ff=0.625

kappa_target = 5.600

kappa_actual = 4.964

beta_N = 0.396 %

growth rate (gamma) = 2.0734E+01 s-1

wall time (tau_w) = -4.8229E-02 s

gamma * tau_w = -1.0000

max coil current = 0.320 MA

Category: passive stable

axis_offset_pct = 4.40%

Equilibrium Statistics

Equilibrium Statistics:	
Topology	= Limited
Toroidal Current [A]	= 1.0186E+06
Current Centroid [m]	= 0.826 0.000
Magnetic Axis [m]	= 0.835 0.000
Elongation	= 4.964 (U: 4.962, L: 4.966)
Triangularity	= -0.649 (U: -0.653, L: -0.645)
Plasma Volume [m^3]	= 6.032
q_0, q_95	= 0.806 13.678
Peak Pressure [Pa]	= 1.2953E+05
Stored Energy [J]	= 9.1278E+04
<Beta_pol> [%]	= 52.8968
<Beta_tor> [%]	= 0.8366
<Beta_n> [%]	= 0.3961
Diamagnetic flux [Wb]	= 1.0241E-02
Toroidal flux [Wb]	= 1.9656E+00
l_i	= 1.3245